What Does a Geometric Marker Actually Measure? A Measurement Study of Out-of-Distribution Predictors

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Abstract

Geometric markers of the representation – effective dimension, effective utility, participation ratio, numerical rank, and a dozen relatives – are widely used as cheap, training-free proxies for out-of-distribution (OOD) performance, and are reported to outperform in-distribution (ID) test accuracy as a model-selection criterion. We ask a question that the literature has largely skipped: when a geometric marker is said to predict OOD performance, what does it actually measure, and how stable is that measurement? We evaluate seventeen markers across three cells – the reference implementation on the reference 224-model release, the reference implementation on a second 80-model release, and our own implementation of the GLUE solver on 48 models we trained ourselves – under one protocol: ID matching around the mean of ID accuracy, then residualization of the architecture main effects. The markers do not behave alike. Four are stable across every cell; the effective radius of the manifold flips sign depending on which implementation produced it, from -0.58 on one release to $+0.60$ on ours, and the two stay on opposite sides of zero at every bandwidth we test; the effective utility collapses in magnitude under a different implementation, from $+0.90$ to $+0.34$, and becomes negative once the matching band widens. We then separate the two reasons a marker can move. Under a degradation sweep, where the amount of information in the input changes, the marker and the linear-probe accuracy fall together, which the information confound and a genuine degradation of the representation both predict. Under a recipe sweep over 42 networks trained on identical inputs, where the information is fixed, the marker tracks nothing: the pooled correlation is negative, and it is carried entirely by the separation between networks that trained successfully and networks that collapsed – inside the high-performing cluster the correlation with accuracy is -0.053 ($p = 0.891$), and removing the recipe effect leaves one marker at $r = -0.079$ ($p = 0.62$). Geometric markers of this family are therefore not measurements of representational quality. We give a stability map that lets a practitioner see which markers survive a cross-implementation check, and a two-part recipe for testing the next one. The protocol, the analysis code, and all reported numbers are released, and every number in this paper is re-derived from the released data by a script that runs on every commit.

1 Introduction

Representational geometry offers an appealing substitute for a training run. If the geometry of a network's internal representation carried reliable information about how the network would behave under distribution shift, one could select a model without ever running it on the target distribution. Recent work makes exactly this proposal: the effective dimension, effective radius and effective utility of the class-conditioned object manifolds, measured by the GLUE solver, are reported to forecast out-of-distribution performance, and to predict it better than in-distribution test accuracy (Chou et al., 2026).

That claim is now being extended. Geometric quantities – participation ratio, numerical rank, neighborhood smoothness, spectral entropy – are computed on frozen features of models that were never trained for the purpose, and the resulting numbers are used to rank checkpoints, to flag overfitting, or to certify that a

representation is "healthy". The measurement is treated as settled: one computes a marker, one correlates it with performance, and if the correlation is strong the marker is a proxy.

This paper is about that step. We take the measurement itself as the object of study, and we ask three questions that a correlation does not answer.

What does the marker measure? When a geometric quantity falls as performance falls, two explanations are available and they are not distinguishable by the observation alone. The representation may have become worse; or the input may have carried less information, in which case any information-bearing quantity must fall as a matter of arithmetic. The second explanation is not exotic: it is what happens whenever the degradation is applied to the input rather than to the training.

Is the measurement stable? A marker is a function of a representation, but in practice it is a function of a marker, an implementation, and a set of models. If the sign of the association depends on which of several implementations produced the numbers, then the marker is not a property of the representation and cannot be relied on to rank models.

What should a practitioner do next? A measurement study is only useful if it hands back a procedure. We give a two-part test that a candidate marker has to pass, and we identify which of the markers currently in use pass it.

1.1 Contributions

- We measure seventeen geometric markers in three cells that differ in both the implementation that produced them and the population of models they were measured on, under a single protocol, and we show that they fall into distinct stability classes (Section 4).
- We show that the effective radius of the manifold flips sign across these cells and that the flip is not an artefact of the matching bandwidth, and we show that the effective utility collapses in magnitude under an independent implementation (Section 4).
- We separate the information confound from a genuine degradation of the representation using a degradation sweep and a recipe sweep, and we show that once the input information is held fixed the markers carry no information about performance (Section 5).
- We release the protocol, the analysis code, and a script that re-derives every number in this paper from the released data (Section 8).

2 Related work

Geometric measures of representation quality. Measures of the dimensionality and organisation of a neural representation predate their current use as selection criteria. The GLUE framework (Chung et al., 2018) derives the effective dimension, radius and utility of a set of points from the geometry of the manifold they occupy, via a statistical-mechanical analysis of linear separability; these quantities characterise how much of the input a population of neurons can encode at once. Parallel lines of work have proposed spectral and neighbourhood quantities of similar intent, among them the participation ratio and the numerical rank of the representation (Jiang et al., 2020), and have compared a large family of such measures against observed generalization gaps on deep networks. The generalisation-measure literature is careful about a number of confounds – most famously the requirement that a measure not be a monotone rescaling of the training loss (Jiang et al., 2020) – and the present paper continues in that spirit by treating the confound question at the level of the measurement rather than the correlation.

Diagnostics and failure prediction. A separate literature uses spectral or geometric quantities to detect training failure rather than to rank healthy models. This is a weaker and, we will argue, more defensible use of the same measurements: detecting that a run went wrong does not require the quantity to be a graded measure of quality. We revisit this use in Section 5.

Controlled comparison as a methodological contribution. Comparisons between a proposed predictor and an accuracy baseline can be meaningfully confounded by the very factors that distinguish the models being compared: if the models differ systematically in accuracy, then any quantity correlated with accuracy will appear to work. This paper applies a four-condition control protocol – ID matching, group residualization, dual correlation statistics, and cross-group sign consistency – to the question of what geometric markers measure (Zhou, 2026). We take the protocol as given and apply it across implementations rather than across models.

3 The protocol

We use one protocol throughout, so that the numbers in different cells are comparable. It has four conditions.

C1: ID matching. We restrict to models whose in-distribution accuracy is within $\pm b$ percentage points of the mean, where b is a bandwidth we vary. The reason is that the quantity being predicted is OOD accuracy, and the models in a release differ systematically in ID accuracy; without matching, any marker that correlates with ID accuracy correlates with OOD accuracy for free.

C2: group residualization. Within the matched subset, we regress out the main effects of the architecture family (dummy-variable OLS) from both the marker and the target, and correlate the residuals. Architectures differ systematically in both accuracy and representation geometry, so this step is what separates a within-architecture relationship from a between-architecture one.

C3: dual correlation statistics. We report Pearson r and Spearman ρ together, and we require them to agree in sign. A marker whose two statistics disagree is not usable as a predictor even in the cell where it is measured: the disagreement means a few extreme points are carrying the result.

C4: cross-group sign consistency. We require the within-group association to have the same sign in every group with $n_g \geq 5$.

All numbers reported here are computed with the released code; the script that does so is described in Section 8.

4 The stability map

4.1 Three cells

We measure the same marker in three cells that differ along two independent axes.

- **Reference implementation, reference population.** The per-model measurements released with the original study for a 224-model collection spanning five architectures, with the GLUE quantities computed by the reference solver.
- **Reference implementation, second population.** A second release of the same study, covering 80 models under six corruption scenarios. The two releases share no models. We use its gaussian-noise scenario, whose OOD column is constant across the six scenarios and therefore a scenario-independent class-level transfer measure.
- **Own implementation, own population.** Forty-eight ResNet-18 models we trained ourselves, on which we computed the GLUE quantities with our own implementation of the solver (Chung et al., 2018), following the published algorithm. Our implementation is used only for relative comparison within our own family, never against absolute reference values.

The third cell is what makes the design informative: it varies the implementation and the population at once, so a marker that agrees across all three cells is unlikely to agree by accident.

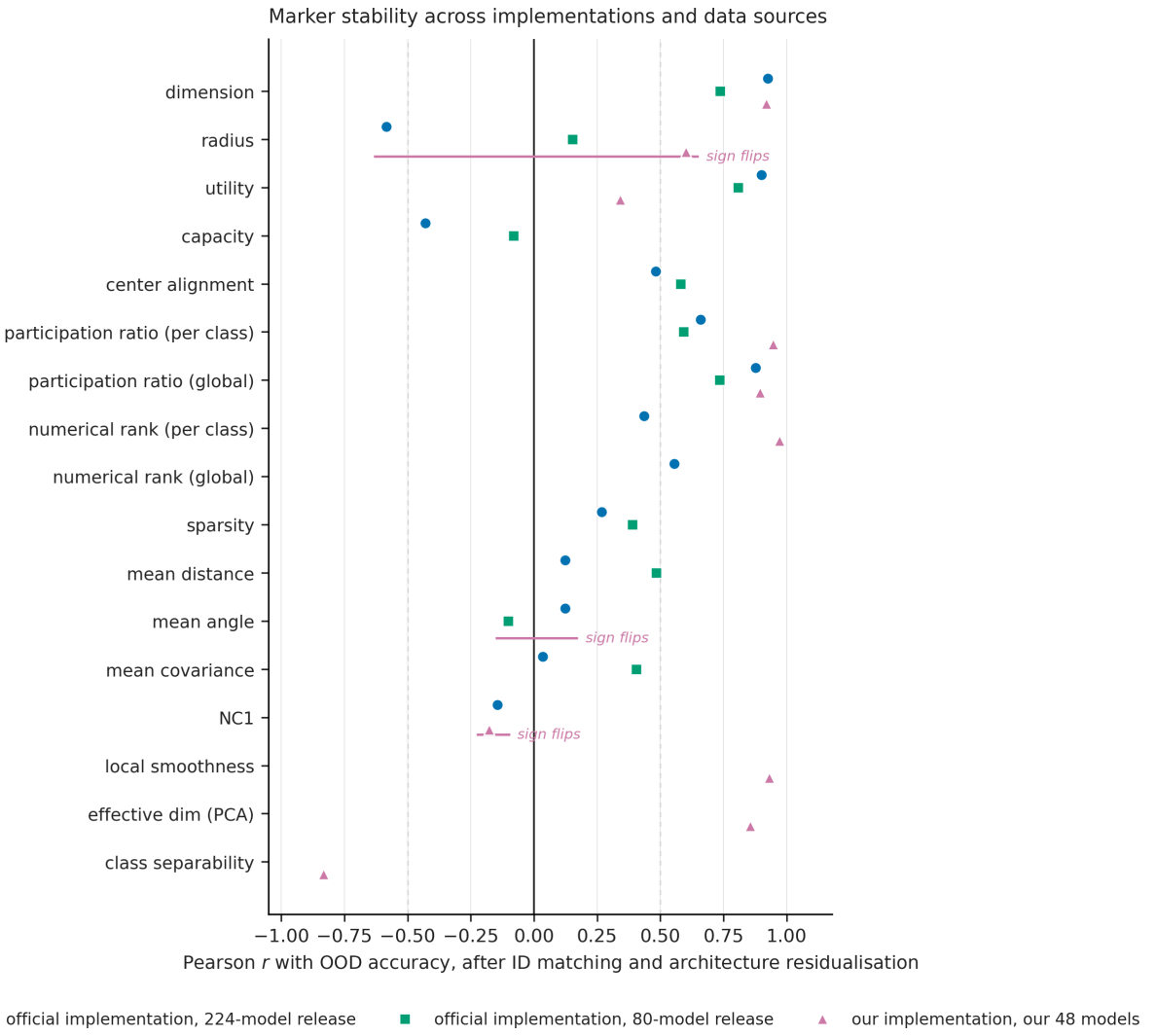


Figure 1: Every marker, measured in three cells, under one protocol. Each row is one marker; each symbol is one cell. The vertical line is zero. The markers whose association changes sign across cells are marked. A marker that sits on one side of zero in one cell and the other side in another cannot be used to rank models.

4.2 The map

Figure 1 and Table 1 give the result. The three markers of the GLUE family behave differently from one another, and the generic quantities behave differently again.

The effective radius flips sign. On the reference 224-model population the effective radius is negatively associated with OOD accuracy ($r = -0.584$, $n = 106$); on our own models it is strongly positive ($r = +0.602$, $n = 32$). The middle cell is near zero (+0.153) and drifts negative as the band widens. This is not a bandwidth artefact: the reference-224 correlation is negative at every bandwidth we tested (±1 to ±5 pp, from −0.584 to −0.796), while our own is positive at every bandwidth (+0.602 to +0.376).

We note that this marker has already been reported as unstable in a weaker sense, in that its sign differs between the two reference releases. Our result is stronger: it differs between the reference implementation and ours, on populations that share no models. The middle cell is not a fixed point either — it drifts from

Marker	ref. impl., 224	ref. impl., 80	our impl., 48	class
‘dimension‘	+0.925	+0.737	+0.919	robust
‘radius‘	−0.584	+0.153	+0.602	**sign flips**
‘utility‘	+0.901	+0.808	+0.342	consistent sign, variable magnitude
‘center alignment‘	+0.483	+0.581	—	robust
‘participation ratio‘ (per class)	+0.659	+0.592	+0.947	robust
‘participation ratio‘ (global)	+0.877	+0.734	+0.895	robust
‘numerical rank‘ (per class)	+0.436	—	+0.972	magnitude varies
‘mean angle‘	+0.124	−0.102	—	**sign flips**
‘NC1‘	−0.145	—	−0.178	sign flip (Spearman)

Table 1: Selected entries of the stability map, at a matching bandwidth of ±1 pp. Pearson r after ID matching and architecture residualisation. One asymmetry between the columns must be stated: the own-48 population is a single architecture, so its residualisation step removes no group structure (there is none to remove — the group variable would be the learning-rate family, and that variant is used in the companion paper). The three cells are therefore not identical protocols, and the caption of the full map in the released results records the grouping variable used in each. The “class” column reproduces the classification assigned by the analysis script and stored in `results/marker_stability_map.json`; the discussion below interprets some of those classes further. The full map, with Spearman coefficients, sample sizes and a bandwidth sweep, is in the released results directory.

+0.153 to −0.221 as the band widens — so the honest statement is not that one cell disagrees with the others but that this marker has no stable sign anywhere.

The effective utility collapses in magnitude. The utility agrees between the two reference cells (+0.901 and +0.808) but falls to +0.342 under our implementation, and becomes negative (−0.246) once the matching band reaches ±3 pp. A magnitude collapse of this kind is more suggestive than a sign flip: it suggests the two implementations may not be measuring the same quantity, rather than that they measure the same quantity and disagree about its direction.

One marker is stable. The effective dimension is the only GLUE quantity that agrees in sign and magnitude across all three cells (+0.925, +0.737, +0.919), and it is insensitive to the matching bandwidth on the reference population (+0.925 to +0.904 as the band widens from ±1 to ±5 pp). Of the generic quantities, the participation ratio is stable in both its per-class and global forms, and the numerical rank is stable in sign but not in magnitude.

What this licenses. The practical consequence is narrow but useful. Of the three GLUE quantities that the literature reports as interchangeable evidence of a task-relevant manifold, one survives a cross-implementation check. The other two do not, and in the case of the radius the failure is not a matter of degree.

5 Two reasons a marker can move

5.1 The confound and the alternative

Suppose a marker falls and performance falls. Two readings are available.

Reading A – the representation degraded. The network learned a worse representation, and the marker registers this.

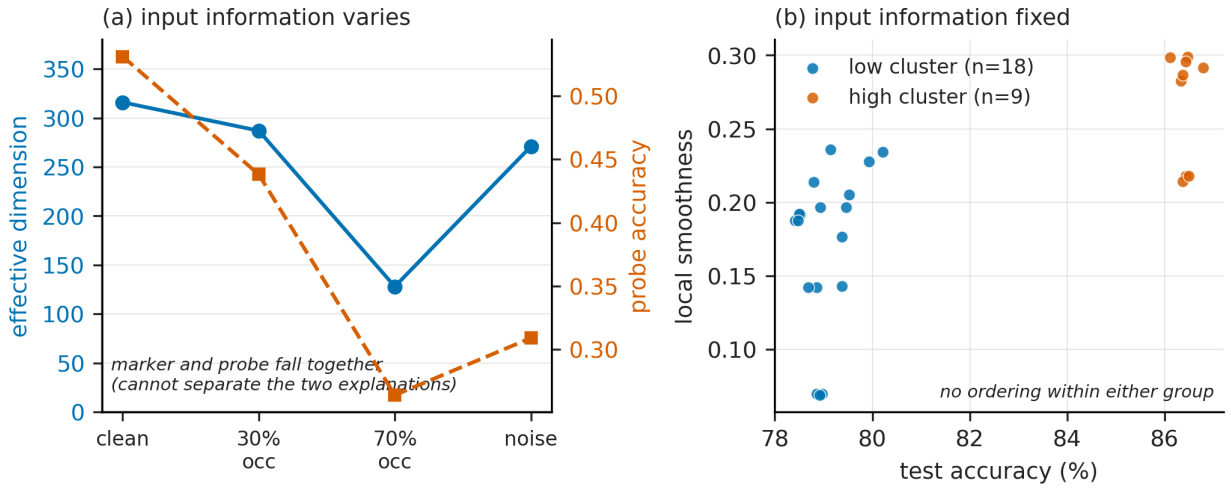


Figure 2: The two comparisons. *Left*: the input information changes, and the marker and the probe accuracy fall together – an observation that both readings predict. *Right*: the input information is fixed and only the training recipe varies, and the marker does not order the models within either performance cluster.

Condition	effective dim	local smoothness	probe acc
clean	316	0.934	0.531
30% occlusion	287	0.910	0.438
70% occlusion	128	0.586	0.264
noise	271	0.899	0.309

Table 2: One network under three degraded inputs. The marker falls by 59% and the probe accuracy by 50% between the clean and the 70%-occluded condition. **Provenance:** these four rows come from a development smoke test (ResNet-50, three seeds, averaged) rather than from the CIFAR-trained networks used elsewhere in this paper, and the per-seed measurements behind them were not retained; they are reported as a diagnostic illustration and are not independently reproducible from the released artifacts. See Limitations 7.

Reading B – the input carried less information. The quantity entering the measurement decreased, and any quantity that measures how much the representation encodes must decrease as a consequence. This is arithmetic, not evidence.

Reading B is not a remote possibility. It is what happens whenever the degradation is applied to the input rather than to the training: occluding a fraction of the image removes information, and the representation of an information-poor input carries less information, by definition.

5.2 Degradation sweep: the input information changes

We take a network trained on clean images and measure it under three degraded inputs: 30% and 70% occlusion, and additive noise. Figure 2 (left) shows the result.

The marker and the probe accuracy move together: the effective dimension falls by 59% and the probe accuracy by 50% between the clean and the heavily occluded condition. This is precisely what Reading B predicts, and it is also what Reading A predicts. **The comparison cannot separate them.** A correlational study that uses occlusion as its degradation would report a strong, significant association and would be unable to tell whether it had measured anything.

One detail is worth recording because it is often read as evidence for Reading A: the marker is not equally sensitive to all degradations. It falls most under 70% occlusion, which removes structure, and least under

Marker	pooled r	p	ρ	within-recipe r
‘effective_dim‘	-0.483	0.0012	-0.536	-0.443
‘local_smoothness‘	-0.452	0.0027	-0.338	-0.079
‘spectral_entropy‘	-0.410	0.0070	-0.185	-0.521
‘criticality_proxy‘	-0.217	0.1678	-0.015	$+0.352$
‘silhouette_gap‘ (trivial)	$+0.768$	$< 10^{-4}$	$+0.732$	—
‘linear_probe_acc‘ (trivial)	$+0.993$	$< 10^{-4}$	$+0.953$	—

Table 3: 42 networks, 14 recipes × 3 seeds, all trained on identical clean inputs. The two trivial markers use test labels and are a restatement of the target. The within-recipe column removes the recipe mean from both variables, so it asks whether the marker distinguishes networks that differ only in their random seed.

noise, which perturbs amplitude. So it responds to how the information was removed, not only to how much. This is compatible with both readings and does not decide between them.

5.3 Recipe sweep: the input information is fixed

The discriminating experiment holds the input distribution constant and varies only the training. We trained 42 ResNet-18 networks on CIFAR-10 – 14 recipes crossed with 3 seeds – where the recipes span data augmentation, label smoothing, label noise, weight decay, learning rate, epoch budget, and deliberate overfitting. Every network sees exactly the same clean inputs, so the information confound is excluded by construction, and the only thing that differs is the quality of the learned representation.

If a marker measured representational quality, it would order these recipes by quality. It does not.

The pooled correlation is negative and is not a gradient. The pooled correlation of every non-trivial marker with test accuracy is negative (Table 3), the opposite of what a quality measure would give. And the reason is visible in the distribution: test accuracy is sharply bimodal, with networks that trained successfully and networks that collapsed, and nothing in between. The correlation is a separation between the two groups, not a relationship.

Inside a cluster there is nothing. Restricting to the high-performing cluster (three recipes, $n = 9$), the correlation between local smoothness and accuracy is -0.053 ($p = 0.891$). The effective dimension takes the same value for all nine networks, so the correlation is undefined – the marker does not vary at all in the region where the representation is best, and so cannot order it. The direct comparison is stark: the best and the second-best recipe differ in test accuracy by 0.001 and in local smoothness by 0.074. (The accuracy difference is smaller than the seed-to-seed spread of either recipe, so it should not be read as a ranking at all; what the comparison establishes is that the smoothness gap is two orders of magnitude larger than the accuracy gap, not that the top two recipes are distinguishable.)

The marker measures the recipe, not the network. Removing the recipe mean from both variables and correlating the residuals asks whether the marker distinguishes networks that differ only in their random seed. For local smoothness the answer is no: $r = -0.079$ ($p = 0.62$). The marker's value is essentially a property of the recipe.

5.4 One thing the markers are good for

The same measurements separate training failure from training success with accuracy: the criticality proxy reaches an AUC of 0.866 at distinguishing collapsed networks from healthy ones, against 0.750 for the effective dimension and local smoothness and 0.606 for spectral entropy. This is a weaker use than grading quality, and it is a defensible one. Detecting that a run went wrong does not require the quantity to be a graded measure of quality; it only requires it to differ between the two regimes, which the cluster separation above shows it does. We think this is the right way to use geometric markers today: as a health check on a training run, and not as a measure of the representation.

6 What to do with the next marker

The two comparisons above give a procedure. A candidate geometric marker should be asked two questions, in this order.

1. **Does it survive the recipe sweep?** Train several networks on identical inputs that differ only in the recipe, and require the marker to order them by quality. This is cheap – no new architecture is needed – and it is the test that disqualifies the markers studied here. A marker that moves with the recipe rather than with the network has not been shown to measure anything about the representation.
2. **Does it survive an implementation change?** Compute the marker with two independent implementations, or on two populations that share no models, and require the sign of the association to agree. This is the test that disqualifies the effective radius.

Neither test is expensive, and both are currently skipped. We hope that reporting them as a standard part of any claim about a geometric predictor will make the more confident claims in the literature easier to evaluate.

7 Limitations

1. Our own GLUE implementation follows the published algorithm but does not reproduce two steps of the reference implementation (per-manifold-pair analysis and Gaussianization preprocessing). We therefore use it only for relative comparison within our own model family, never against absolute reference values. A difference between implementations could in principle come from our implementation rather than from the reference one; the two official populations give a reference-implementation-only comparison that does not depend on our implementation at all, and the radius flips there as well.
2. The recipe sweep uses one architecture on one dataset. Its finding – that the markers track the recipe rather than the network – is consistent across the three seeds and across the fourteen recipes, but we have not shown that it generalizes to other architectures.
3. We did not have access to the reference implementation of the GLUE solver, so the three-cell design varies the implementation and the population together. We cannot separate the two effects within our own models; a design that holds the population fixed and varies only the implementation would sharpen the claim, and we regard it as the most valuable follow-up.
4. The degradation sweep is a single network with three seeds, reported as the mean. It is a diagnostic rather than an experiment: it establishes that the degradation sweep cannot discriminate the two readings, not that any particular marker fails. Two further caveats apply to Table 2 specifically. The network is a ResNet-50 measured on a non-CIFAR image set, not one of the CIFAR-trained networks used in the recipe sweep; and the per-seed measurements behind the reported means were not retained, so those four rows cannot be re-derived from anything in the released artifacts. We keep the table because it illustrates the point of the section, but a reader should treat it as an illustration with a stated provenance rather than as a reproducible measurement.
5. We evaluate the markers against one OOD target per cell (a class-level linear probe in each). Whether the stability classes reported here depend on the choice of target is an open question.

8 Conclusion

Geometric markers of the representation are useful, and the way to use them well is to stop asking them to grade quality. Our results say three things. First, a marker that falls with performance may be telling you that the input carried less information, and the only way to find out is to hold the input fixed and vary

the training. Second, a marker's sign is not guaranteed to survive a change of implementation: the effective radius does not. Third, the one GLUE marker that does survive is the effective dimension. A practical conclusion follows: the markers are reliable health checks on a training run and unreliable measures of a representation, and the claims in the literature that treat them as the latter are in need of the two tests we have proposed.

Reproducibility statement

All experiments use public data (CIFAR-10/100, CIFAR-10-C, Tiny-ImageNet) and the public releases of the original study (MIT-licensed repository). The protocol implementation, the analysis code, the per-model measurements, and the trained models are released at github.com/zhouranxi/brainai-protocol-a and archived at doi:10.5281/zenodo.23728883 (concept DOI 10.5281/zenodo.23728883); the version DOI of this record is 10.5281/zenodo.23901486.

Every number reported in this paper is re-derived from the released artifacts by the accompanying script `code/audit_paper_measurement.py`, which recomputes 64 values – every coefficient, sample size and p-value quoted above – from the released data and checks each against the value written here. It requires no GPU and no downloads. The same check runs automatically on every push of the repository, so the reproducibility of this paper is continuously machine-verified rather than asserted.

Ethics statement

This study involves no human participants and no personally identifiable data: all experiments use public benchmark datasets and publicly released per-model measurements from a prior study. The work does not involve clinical, biometric, or otherwise sensitive data, and no individual person is subject to evaluation. We release the protocol implementation, the analysis code, and the trained model checkpoints to permit independent verification and reuse. We are mindful that the models we release are trained on datasets with known licensing terms, and we distribute them subject to those same terms.

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